DECODELABS – DATA ANALYTICS PROJECT 2

Retail Sales Forensics & Customer Intelligence

Author: Hijab Zahra Khan **Topic:** Exploratory Data Analysis (EDA) **Status:** Completed

📌 PROJECT OVERVIEW

This project focuses on Exploratory Data Analysis (EDA) using Python to investigate retail sales data, identify data quality issues, discover patterns, and generate meaningful business insights. The project demonstrates how raw data can be transformed into clean, structured, and useful information to support business decision-making.

> **Note:** This project uses a simulated retail sales dataset. The findings are illustrative and do not represent actual company sales.

🎯 PROJECT OBJECTIVES

- Explore and understand the structure of retail sales data.
- Identify missing values, duplicate records, invalid dates, and inconsistent entries.
- Clean and prepare data for analysis.
- Calculate descriptive statistics, including mean, median, and count.
- Analyze distributions and identify potential outliers.
- Apply IQR and Z-score methods for outlier detection.
- Explore relationships between numerical variables using Pearson correlation.
- Analyze monthly sales trends, product categories, regions, and customer segments.
- Present findings through data visualizations.
- Develop actionable business recommendations.

🛠️ TECHNOLOGIES & LIBRARIES

Tool / Library	Purpose
Python	Programming and analysis
Pandas	Data manipulation & cleaning
NumPy	Numerical operations
Matplotlib	Data visualization
Openpyxl	Excel file export

📂 DATASET OVERVIEW

The simulated dataset contains approximately **1,500 retail transactions** with intentionally introduced data quality issues to demonstrate data forensics and cleaning techniques.

> **Revenue Calculation:**
> **Revenue = Quantity × Unit_Price**

📋 DATASET SCHEMA & ATTRIBUTES

Column	Description
Transaction_ID	Unique transaction identifier
Date	Transaction date
Customer_ID	Customer identifier
Category	Product category
Region	Sales region
Quantity	Number of items purchased
Unit_Price	Price per item
Payment_Method	Payment method
Revenue	Total transaction revenue

🔍 PROJECT WORKFLOW

1. **Data Generation & Exploration:** Examined dataset dimensions, data types, missing values, duplicate records, and initial descriptive statistics.
2. **Data Cleaning & Preparation:** Handled missing values, invalid dates, duplicate records, negative quantities/prices, applied median imputation, recalculated revenue, and extracted temporal fields (month, quarter).
3. **Statistical Analysis:** Evaluated mean, median, count, standard deviation, quartiles, five-number summary, distribution, and skewness.
4. **Outlier Detection:** Executed Interquartile Range (IQR) and Z-score methods to flag unusual observations.
5. **Correlation Analysis:** Calculated Pearson correlation coefficients between quantity, unit price, and revenue, evaluating correlation versus causation.
6. **Business Analysis:** Evaluated monthly revenue trends, product category performance, regional distribution, payment methods, and customer spending segments.

📈 DATA VISUALIZATIONS

- **Monthly Revenue Trend:** Line Chart
- **Category Performance:** Bar Chart
- **Revenue Distribution:** Histogram
- **Revenue Outliers:** Boxplot
- **Correlation Analysis:** Heatmap
- **Regional Sales Performance:** Bar Chart
- **Customer Segmentation:** Bar Chart

💡 BUSINESS INSIGHTS

- Monitor monthly revenue patterns and seasonality.
- Identify high-performing vs. lagging product categories.
- Compare regional sales performance to target growth.
- Understand customer spending behavior and tiers.
- Investigate unusual or fraudulent transactions.
- Ensure data integrity prior to strategic decision-making.

📁 PROJECT STRUCTURE

```
Retail-Sales-Forensics/ | ├── README.md ├── retail_sales_eda.ipynb | └──
DecodeLabs_Project_2/ ├── raw_retail_sales.csv ├── clean_retail_sales.csv ├──
retail_sales_cleaning_report.csv ├── retail_sales_outliers.csv ├──
retail_sales_analysis.xlsx | └── charts/ ├── monthly_revenue.png ├──
category_performance.png ├── revenue_distribution.png ├── revenue_boxplot.png ├──
correlation_heatmap.png ├── regional_sales.png └── customer_segments.png
```

▶️ EXECUTION GUIDE

1. **Clone Repository:** `git clone YOUR_GITHUB_REPOSITORY_URL`
2. **Install Dependencies:** `pip install pandas numpy matplotlib openpyxl`
3. **Launch Notebook:** Open `retail_sales_eda.ipynb` in Jupyter Notebook or Google Colab.
4. **Run Pipeline:** Execute cells sequentially to run data generation, cleaning, analysis, and plot output.

📚 KEY LEARNING OUTCOMES & CONCLUSION

Through this project, proficiency was demonstrated in Exploratory Data Analysis (EDA), data preprocessing, outlier detection (IQR & Z-score), Pearson correlation, visualization, and strategic insight generation. This project underscores the end-to-end EDA lifecycle, reinforcing the vital role of data integrity, analytical reasoning, and clear visual presentation in converting raw datasets into executive intelligence.